GH raisoni INTERNATIONAL SKILL TECH UNIVERSITY
Pune

AI CHATBOT FOR COLLEGE

FACULTY REVISED SYSTEM ARCHITECTURE & DFD SPECIFICATION

✓ Context-Level DFD (Level 0) Included ✓ Standard Open-Ended Data Stores ✓ Circle Process Bubbles (DeMarco/Gane-Sarson)

Data Flow Diagram (Level 0: Context-Level System Boundary)

Context-Level Data Flow Diagram (DFD Level 0)

High-Level System Boundary & External Entity Information Flows



Academic Context-Level DFD Design Rules & Justification

1. Single Process Rule (Process 0.0): In Level 0 (Context Diagram), the entire software system is abstracted into exactly ONE central circle process.
No internal subprocesses or data stores are revealed at Level 0, establishing clear architectural boundaries.

2. External Entities (Squares): Represents external actors that exchange information with the chatbot: Student and Administrator.

3. Balanced Data Flows: All data entering and leaving the system boundary must match the external flows in the Level 1 DFD.

4. Project Grounding: Covers Pune Wagholi & Yerawada campus inquiries, attendance checks against the official 4-tier detention criteria, real-time speech input/output, and administrative CMS operations.

Fig: Context-Level Data Flow Diagram (DFD Level 0) — High-level system boundary encapsulating the chatbot system as a single central circle process 0.0 with external entities Student and Administrator.